

BRAXTON VOGEL

Software Engineering Student

Spring, TX | 346-413-7560 | bgv007@shsu.edu | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

SUMMARY

Software Engineering student (B.S., expected Dec. 2027, Major GPA 3.6) with hands-on full-stack, cloud, and applied-ML experience across seven independently built and deployed systems. Shipped a production ASP.NET Core API to Azure with a working CI/CD pipeline, a native desktop AI application with multi-provider LLM routing and hardened authentication, and a machine-learning system with ~83% predictive accuracy. Comfortable owning a project end to end — architecture, implementation, deployment, and debugging in production.

EDUCATION

Sam Houston State University

Bachelor of Science in Software Engineering

Major GPA: 3.6 | Dean's List: Fall 2025, Spring 2026

Huntsville, TX

Expected December 2027

Certification: Google Data Analytics Professional Certificate — SQL, Data Cleaning & Transformation, Data Visualization, Spreadsheets, R

TECHNICAL SKILLS

Languages: C#, Python, Java, TypeScript, Rust, SQL, HTML/CSS

Frameworks & Libraries: ASP.NET Core, Entity Framework Core, Next.js, React, Spring Boot, Streamlit, Tauri

Cloud, DevOps & Tools: Azure App Service, GitHub Actions (CI/CD), Redis (Upstash), Vercel, Git/GitHub, REST APIs

Machine Learning & Data: Scikit-learn, Pandas, Random Forest, SHAP, Feature Engineering, Tableau

Concepts: Onion/Clean Architecture, MVC, JWT/PBKDF2 Authentication, Database Design (SQL/ERD), Multithreading & Sockets

PROJECTS

TaskFlow — Task/Project Tracker API

C#, ASP.NET Core, Entity Framework Core, Azure App Service, GitHub Actions

- Built and deployed a production ASP.NET Core Web API using **Onion Architecture**, with four strictly inward-dependent layers (Domain, Application, Infrastructure, API)
- Enforced core business rules (task-state machine) at the domain layer and proved them with a unit test suite backed by an in-memory repository, decoupled from the database
- Deployed a live CI/CD pipeline to Azure App Service via GitHub Actions and diagnosed real production issues, including Conditional Access policy conflicts and resource quota exhaustion

SammyOS — Native Desktop AI Workspace

Tauri, Rust, Next.js, TypeScript, Python, Redis (Upstash)

- Built a native Windows desktop app (Tauri + Rust + Next.js) unifying real-time screen capture, live audio transcription, file/codebase analysis, and autonomous web research into one AI assistant
- Designed a **multi-provider LLM fallback router** across 5 providers (Ollama, user-supplied keys, Groq, Gemini, Cerebras) for near-zero downtime
- Implemented hardened authentication (PBKDF2, Redis-verified sessions) with IP- and account-level rate limiting, plus a distributed async job pipeline for file analysis
- Shipped a live telemetry dashboard and a self-updating silent launcher (Python + Windows Task Scheduler)

Student Risk Prediction System — ML Web Application

Python, Scikit-learn, Pandas, SHAP, Streamlit

- Trained a Random Forest classifier on the Open University Learning Analytics Dataset, reaching **~83% accuracy (ROC-AUC 0.834)** on held-out data
- Built a SHAP-based explainability layer surfacing each student's individual top risk factors and rule-based intervention suggestions
- Deployed an interactive Streamlit app with CSV upload, live inference, and a local LLM assistant (Llama 3 via Ollama) for plain-language explanations of predictions

Additional Projects

Stateful Browser Automation Engine (Python, Playwright) — persistent authenticated-session automation framework. **D&D Character Builder** (Java, Spring Boot, SQL) — normalized schema with many-to-many entity modeling. **Chat Server** (Java, Sockets) — multithreaded, thread-per-client chatroom server. **File Manager UI** (Vanilla JS) — in-memory virtual filesystem with recursive path resolution. **Portfolio Website** (Next.js, TypeScript) — dynamic project routing, GitHub+Vercel CI/CD.

Source code and full write-ups: [GitHub](#) | [Portfolio](#)

LEADERSHIP & COMMUNITY

Community Volunteer — food distribution for underserved communities (local church outreach). **Veteran Support** — Wreaths Across America and flag placement events. **Fundraising** — nonprofit fundraising with local Jeep Club (Noah's Ark Foundation events, car shows, raffles).